

Jeffrey Li

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Education

Worcester Polytechnic Institute (WPI)

Worcester, MA

B.S and M.S in Computer Science - Graduated with High Distinction

May 2026

Relevant Coursework: Software Engineering, Data Structures and Algorithms, Object-Oriented Analysis and Design, Machine Learning, Database Systems, Systems Programming

Work Experience

Ingredion Incorporated, *Software Engineering Intern*

May 2025 – August 2025

- Designed and implemented a LabVIEW application for real-time defect detection in starch filtration machinery.
- Built and deployed a real-time audio classification pipeline using PyTorch and a 1D CNN to detect mechanical anomalies.
- Developed a signal processing pipeline using SciPy and Librosa (STFT, Mel spectrograms, MFCCs) for feature extraction from audio data.
- Implemented data preprocessing and feature engineering using NumPy and Pandas, achieving 98% accuracy on test data.

Acton Institute of Computer Science, *Full-Stack Developer*

June 2023 – August 2023

- Developed a full-stack web application for summer camp registration using React, Express, and Firebase.
- Implemented user authentication, registration workflow, and data persistence for 130+ user accounts.
- Collaborated in a team of five in an Agile (Scrum) environment, delivering features through iterative development cycles.

Projects

Personal Project - Flower Order Tracker

- Built an end-to-end full-stack ordering system for a cross country team fundraiser using React and Spring Boot, enabling door-to-door sellers to record orders and track delivery status.
- Implemented role-based authentication (SELLER/ADMIN) with secure API access and seller-scoped order visibility.
- Designed REST APIs and relational schema with MySQL, including backend pricing logic (3-for-\$25 bundles) and validation.

Personal Project - AI Resume Reviewer

- Built an AI-powered resume reviewer that parses PDF resumes, compares them against job descriptions, and generates ATS-style match scores, targeted feedback, and keyword gap analysis.
- Designed a full-stack web application using React/Next.js and Flask with JSON-based APIs to support resume uploads, structured analysis results, and an interactive feedback dashboard.
- Integrated LLM-driven evaluation with rule-based scoring to suggest stronger resume bullet points, highlight missing skills, and improve resume alignment for specific roles.

Software Engineering Course Project - Hospital Application in Collaboration with Brigham and Women's Hospital

- Collaborated in a team of 11 to develop a cloud-based hospital navigation and service request system using React, TypeScript, and PostgreSQL.
- Contributed to frontend architecture and database design supporting real-time pathfinding, voice navigation, and service request automation.
- Implemented authentication using Auth0 and deployed infrastructure using AWS EC2 and RDS.

Skills

- **Languages:** Python, TypeScript, HTML, CSS, Java, JavaScript, SQL, C++, C
- **Frameworks & Tools:** React, Next.js, Spring Boot, Node.js, Express, Docker, Git/GitHub
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Machine Learning:** PyTorch, NumPy, Pandas, SciPy, Librosa, Scikit-learn
- **Other:** RESTful APIs, CI/CD, NextAuth, Auth0

Teaching/Leadership Experience

Worcester Polytechnic Institute, *Peer Learning Assistant*

August 2023 – Present

- Supported instruction for various math courses, leading weekly recitation sections for 25-30 students.
- Conducted office hours, tutored students and graded coursework for 100+ students.
- Collaborated with faculty and peers to enhance course delivery and improve student outcomes